



JEREMIAH NYAGAH NATIONAL POLYTECHNIC

DEPARTMENT: ELECTRICAL AND ELECTRONICS ENGINEERING

SUBJECT: INSTALL ELECTRICAL POWER LINES

CLASS: LEVEL 6, POE EXAM 4

PERIOD: 2 HRS

Instruction:

- a) Attempt **ALL** the **QUESTIONS**
- b) Any form of cheating is not allowed
- c) Phone are not allowed in the exam room

Question 1 (20 marks)

1.

- (a) List FIVE main components of overhead lines. (5 marks)
- (b) List FOUR commonly used conductor materials. (4 Marks)
- (c) List FOUR properties that the conductor material used for transmission and distribution of electric power should have. (4 marks)
- (d) Copper is an ideal material for transmission and distribution of electric power, however, aluminium is widely used for overhead power lines. Explain why. (4 Marks)
- (e) What do you understand by the term “A.C.S.R.” and state two advantages of “A.C.S.R.” conductors. (3 Marks)

Question 2 (20 marks)

2.

- (a) List FIVE properties of line supports. (5 Marks)
- (b) List FIVE types of line supports. (5 Marks)
- (c) List FIVE main objections to wooden supports (5 Marks)
- (d) List FIVE properties of insulators. (5 Marks)

Question 3 (20 marks)

3.

- (a) List FOUR types of insulators (4 Marks)
- (b) Give SIX advantages of using suspension-type insulators over the pin-type insulators. (6 Marks)
- (c) What do you understand by the term “string efficiency” and write the mathematical equation for “string efficiency” (4 Marks)
- (d) State and explain THREE methods of improving string efficiency. (6 Marks)

Question 4 (20 marks)

4.

- (a) Derive the string efficiency mathematical expression (4 Marks)
- (b) Each line of a 3-phase system is suspended by a string of 3 similar insulators. If the voltage across the line unit is 17.5 kV, calculate the line to neutral voltage. Assume that the shunt capacitance between each insulator and earth is 1/8th of the capacitance of the insulator itself. Also find the string efficiency (8 Marks)
- (c) An insulator string consists of three units, each having a safe working voltage of 15 kV. The ratio of self-capacitance to shunt capacitance of each unit is 8 : 1. Find the maximum safe working voltage of the string. Also find the string efficiency (8 Marks).